

CISCO VIRTUAL INTERNSHIP 2026

SmartBranch 360

Secure Branch-Office Network — Individual Summary

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CONTRIBUTION	Member 3 — Python Tool, Documentation & Demo

Project Overview

SmartBranch 360 is a secure branch-office network built in Cisco Packet Tracer for the Cisco Virtual Internship 2026. It covers wired employee access, a separated guest Wi-Fi segment, an internal server, NAT-based internet connectivity, and locked-down administrative access — verified end-to-end by a Python checker script.

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Member 1

Core Network Build

2

Member 2

Security, NAT & Testing

3

Member 3

Python Tool & Docs (You)

My role hardens and verifies the network Members 1 & 2 built, then documents it for evaluation.

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VLANS

5

FAULT SCENARIOS

21

CHECKS RUN

3

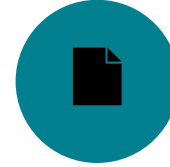
MEMBERS

My Contribution — Member 3



Python Verification Tool

Built smartbranch_checker.py to validate the live config against requirements.yaml



Requirements File

Authored the YAML file capturing VLAN, IP, and security requirements



Design Document

Compiled the topology, VLAN/IP table, and security notes into one report



Fault Cards

Documented the 5 fault scenarios: symptom → root cause → fix



Demo Recording

Recorded the 5–10 minute walkthrough following the project flow

Network Design Summary

VLAN	Name	Network	Gateway
10	Employee	10.10.10.0/24	10.10.10.1
20	Guest	10.10.20.0/24	10.10.20.1
30	Server	10.10.30.0/24	10.10.30.1
99	Management	10.10.99.0/24	10.10.99.1

Design Highlights

- Router-on-a-Stick with per-VLAN sub-interfaces
- 802.1Q trunk (Fa0/1) carrying all 4 VLANs
- 4 DHCP pools, one per VLAN
- Guest VLAN blocked from Server & Management
- NAT overload for shared internet access
- SSH v2 restricted to Management VLAN only

Traffic Flow



Python Verification Tool

`smartbranch_checker.py`

Reads exported router/switch configs and validates them against requirements.yaml — no new network design needed, just automated proof the build is correct.

VLAN Config

4 VLANs exist & active

Trunking

All VLANs on Fa0/1

Gateways

Correct sub-interface IPs

DHCP Pools

Network + default-router

Security

Guest deny rules present

NAT

Overload rule configured

SSH

v2 + Management-only access

Verification Results

21/21

CHECKS PASSED

STATUS: ALL PASSED

Test Area	Checks	Result
VLAN Configuration	4 / 4	PASS
Trunk Configuration	4 / 4	PASS
Router Sub-interfaces	4 / 4	PASS
DHCP Pools	4 / 4	PASS
Security (Guest ACL)	2 / 2	PASS
NAT Overload	1 / 1	PASS
SSH Access Control	2 / 2	PASS

Every check was run against the final exported router and switch configuration files — confirming the build matches the requirements document exactly, with zero manual re-verification needed.

Conclusion

The SmartBranch 360 build satisfies all VLAN, trunking, routing, DHCP, security, NAT, and SSH requirements. The automated checker confirms the configuration is consistent and complete — proving out the network before final submission.

- Full network verified against requirements
- Automated tooling saves manual re-checking
- Documentation ready for evaluation

Thank you